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Executive Director
Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development
Department of Science and Technology

Thru:
Dr. Marissa N. Estrella
Director, BCAARRD

Dear Sir,

Greetings!

This is to respectfully endorse the paper publication entitled "A Framework for Applying Semantic Web in the Design and Development of Web Portals for Agricultural Crops: The Pili (Canarium Ovatum Engl.) Case".

The above is an output of the PCAARRD-funded project entitled "*A Comprehensive PILI (Canarium ovatum Engl.) Information System Utilizing Web 2.0 Technologies: A Roadmap Towards KM Buildup*", which is a component of the program "*Accelerated R&D Program for Capacity Building of Research and Development Institutions and Industrial Competitiveness: Niche Centers in the Regions for R&D (NICER) Program: Pili Research & Development Center*".

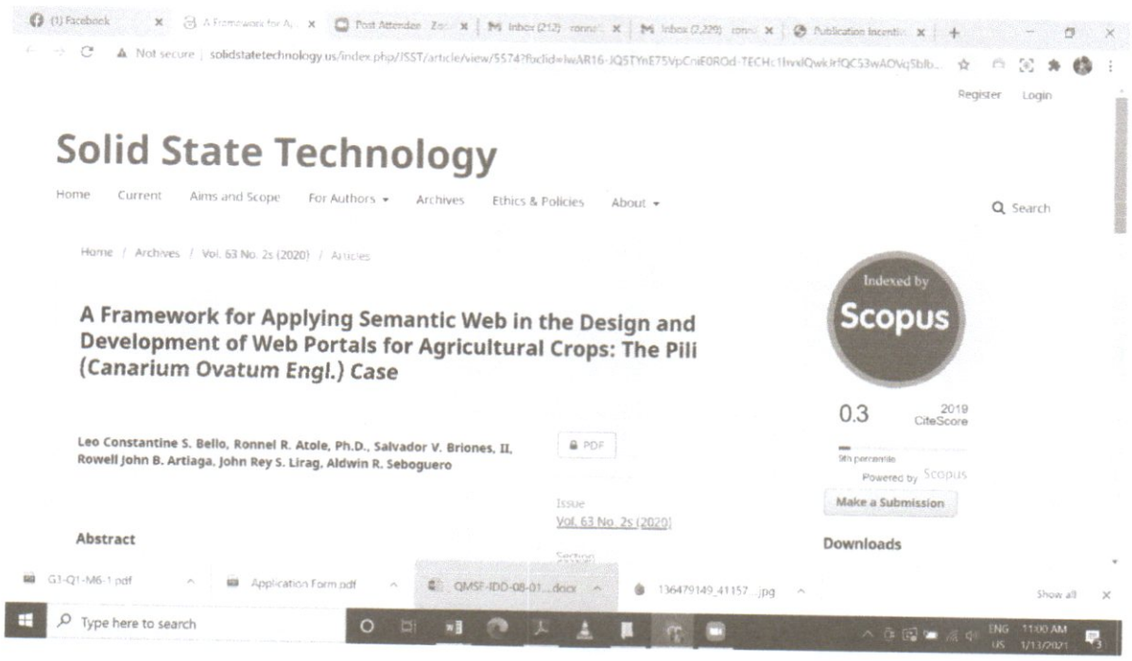
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A Framework for Applying Semantic Web in the Design and Development of Web Portals for Agricultural Crops: The Pili (Canarium Ovatum Engl.) Case

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PDF

Issue
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Abstract

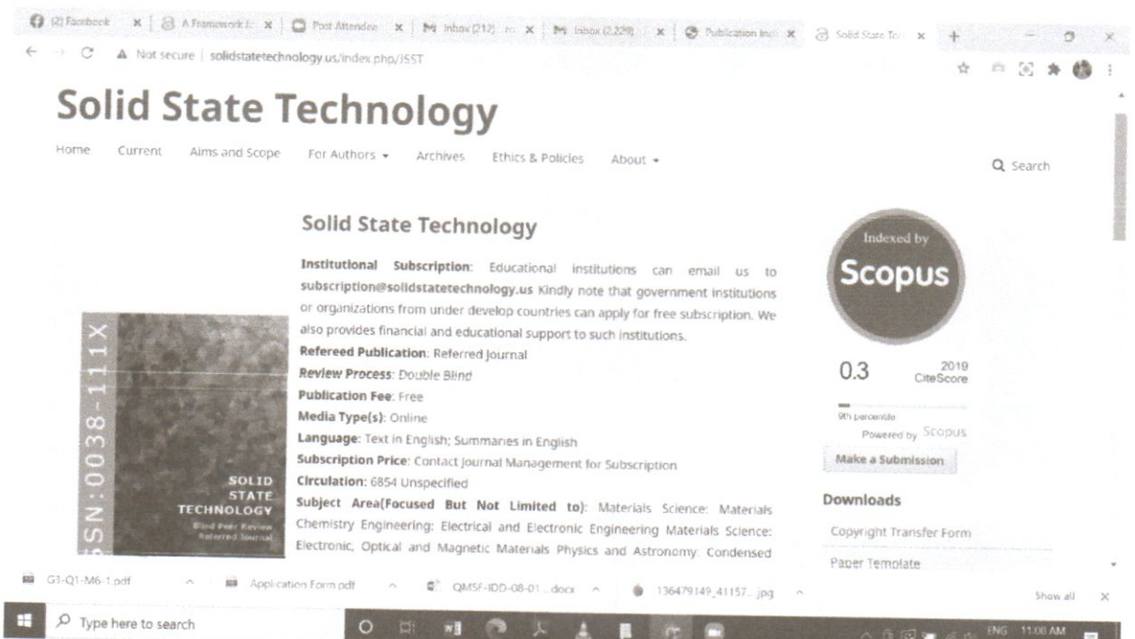
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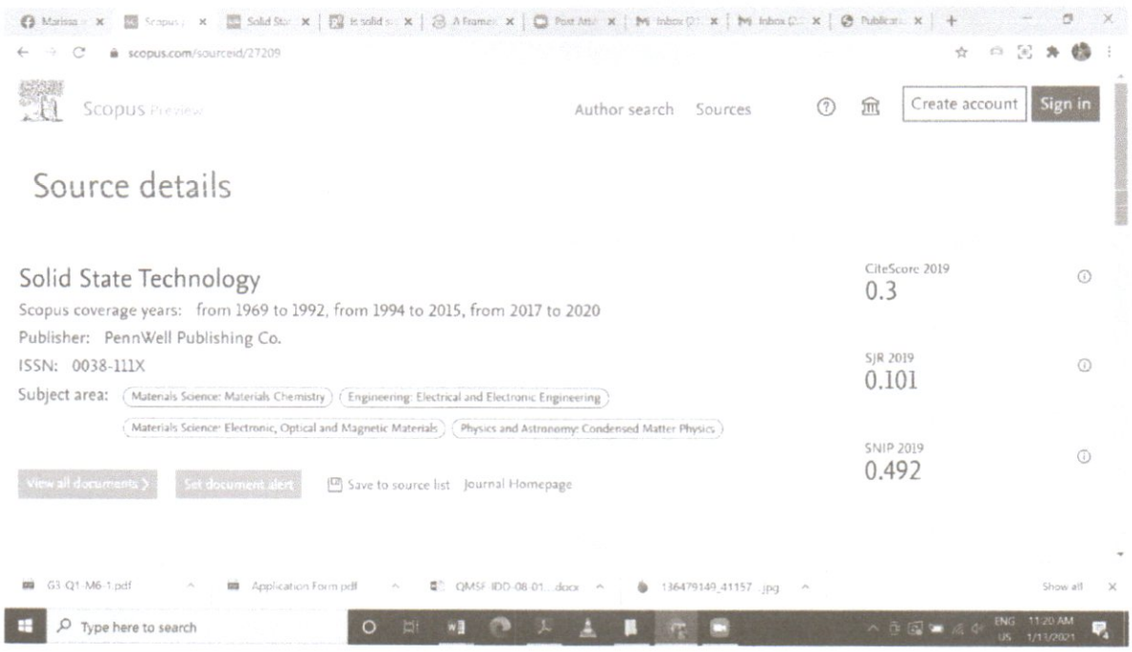
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Figure 5: Condensed Entity-Relationship Model of Pili Knowledge Management System database design

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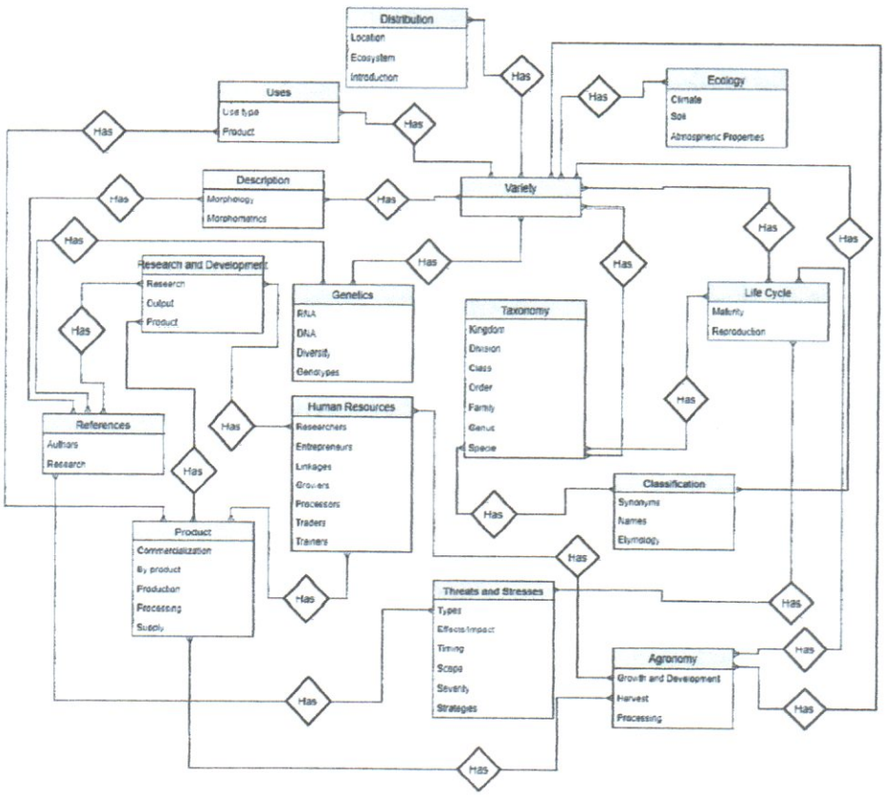
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Pili commodity to be collected in a centralized repository with participants not only to view information but for themselves to share and contribute.

Even though the structure of the Knowledge Base is to collect and consolidate information into one location, it is unlikely that all the information needed by the farmer will be available in the place. This sparks the idea of enabling the Knowledge Management System to communicate with other Knowledge-giving sites. With this, we employed the Semantic Web Technology in designing the Knowledge Management System. To make sure that it is in relevance with the standard Ontologies, we used the structure of the available schemes and standards available in the World Wide Web. We successfully modeled and set-up the structures of the Pili KMS to be Semantic Web-ready once our system is up and is being used by the users and stakeholders. Although the database and its semantic counterpart were already designed and developed, it is expected that it will expand and evolve as needs arise.

FUTURE WORKS

Once the database has been populated and is functional, and the ontologies are well developed, it can now be a valuable resource for many aspects of plant and crop researches. It can now open opportunities for comprehensive data analytics such as comparison of gene expression patterns, comparative morphology, and genetics for the varieties and other species of Canarium Plant among many others.



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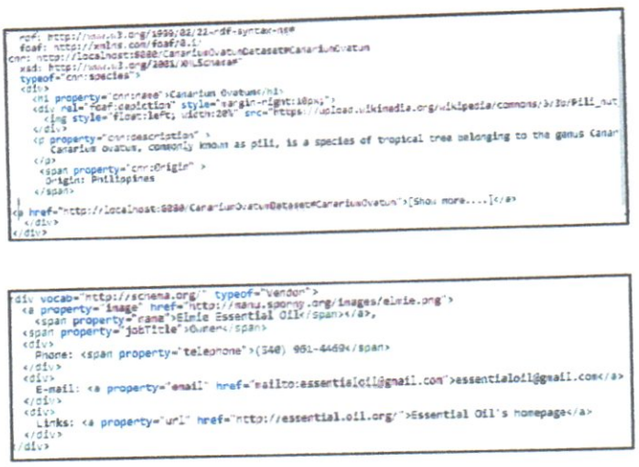


Figure 6: Screenshot of sample HTML Codes implemented with RDF Identifiers (top) Canarium Ovatum properties and (bottom) a sample Human Resource.

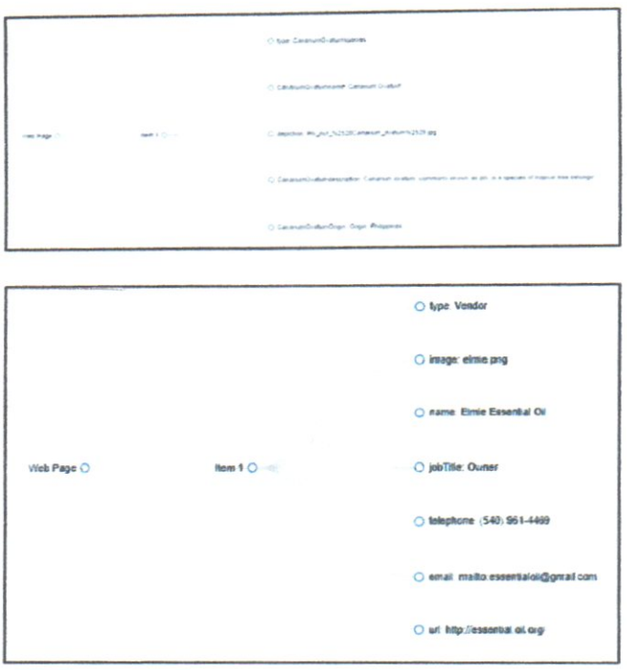


Figure 7: RDF Graph showing (top) Canarium Ovatum properties and (bottom) sample Pili product vendor

4 CONCLUSIONS

Presented in this paper are the outcomes of the design and development of Pili (Canarium Ovatum Engl.) Knowledge Base through the use of Structure Relational Database Model. The knowledge base aims to provide a centralized repository of data in a collaborative environment. Web 2.0 or the Social Web's properties of sharing and build-up of information make it perfect for the knowledge regarding the

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Figure 4: GraphDB interface showing the Class Relationship built from the RDF

3 RESULTS AND DISCUSSIONS

3.1 Database Structure and Design

The database of the Pili Knowledge Management System serves the central repository of data from the collaborative build-up of information.

The data model is built in a structured manner containing a variety of data types including texts, numbers (integers and floating-point numbers), dates, images, and hyperlinks. In Section 2, we discussed the categories of the tables needed to create the repository, namely Basic Information, Promotions, Human Resources, and Research and Development. From the four (4) categories we resulted in fifteen (15) main classes of entities. Fig.5 illustrates a condensed Entity-Relationship Model (for readability) showing the fifteen (15) classes as main entities along with their sub-entities, all are interconnected in many to many cardinalities.

Though this is only an initial structural physical design of the database, we have constructed, using the Third Normal form of Normalization, 152 tables including sub tables. We are expecting that the database will improve and be optimized as inputs from our development team and the stakeholders arrive depending on the need.

3.2 Integration of Semantic Web to the KM Web Structure

To enable the pages for Semantic Web, the HTML codes were embedded with RDF properties as shown in Fig.6. An RDF diagram was generated to visualize the structure of the RDF document embedded in the HTML code (Fig.7).

Through the integration of the Semantic Web to the Knowledge-base, every information that was tagged or marked with RDF Properties will be available to both internal and external entities. This due to the application of the standards provided by OWL used for agriculture establishing commonalities across crops. This will enable the Pili Knowledge-base to access data not present in its database.

A search functionality enabled with semantic-like queries can be accepted and will be queried across all the sematic websites active in the globe providing an interlinked sharing of knowledge.

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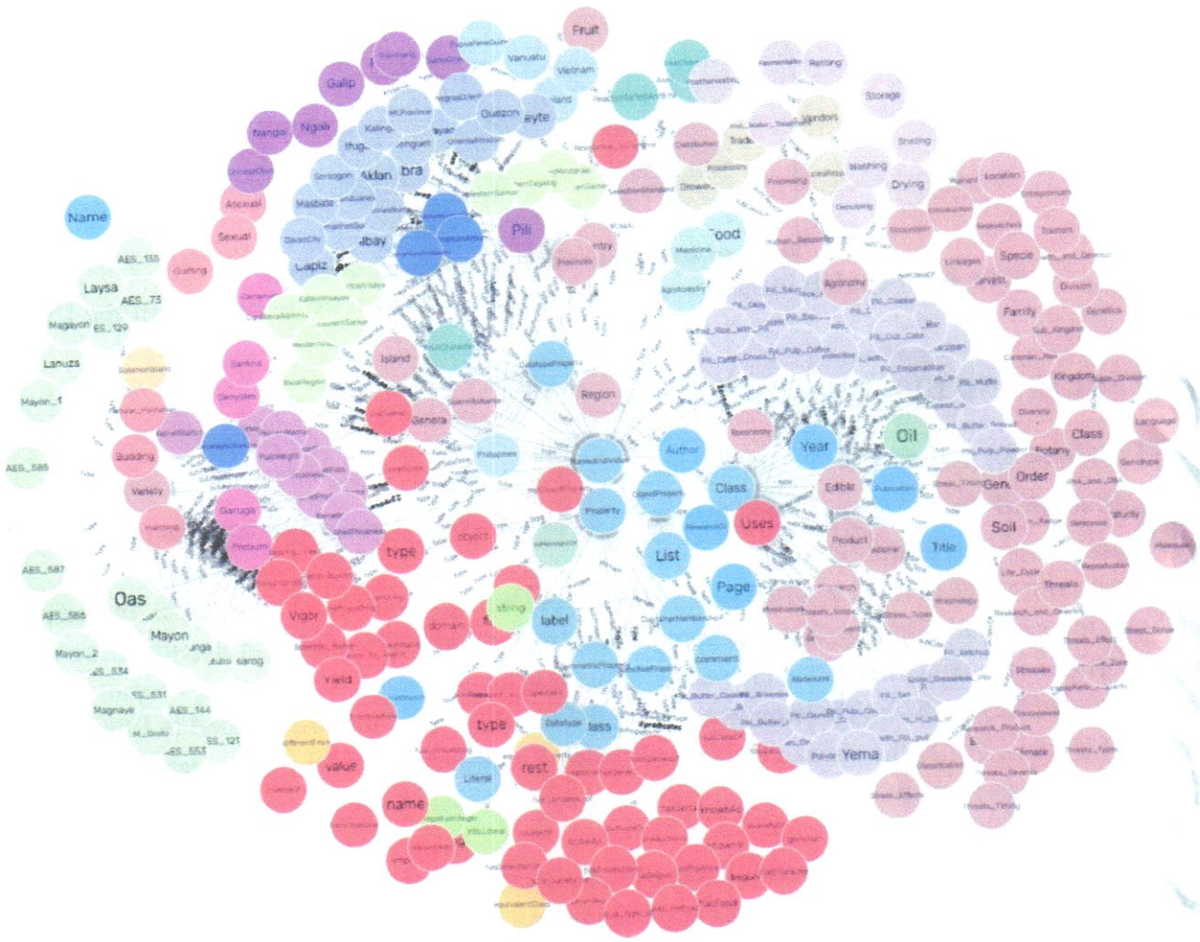


Figure 3: RDF representation of sample sets of facts found in the Pili KMS

2.5 Semantic Web Tools

The following are used to create a representation model and simulate the KMS Semantic Web Setup.

1. *Protégé* – is an open-source ontology and framework used to build a knowledge-based solution in diverse areas [9]. In this paper, is used to generate the RDF file in ".ttl" format.
2. *GraphDB* - an enterprise-ready Semantic Graph Database, compliant with W3C Standards [10]. This was used to visualize the facts in the RDF model (Fig.4).
3. *SPARQL* - an acronym for SPARQL Protocol and RDF Query Language [11], is used to simulate the access of facts and definitions as well as manipulate data stored in the RDF-formatted file.

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1. *Base information*. The base information will be comprised of the Taxonomy, Botanical Description, Properties, Molecular Biology, Habitat and Ecology, and Cultivation and Uses. This will include all the information needed to understand Pili when it comes to its physical properties and uses.
2. *Promotions*. This will include information needed to commercialize and promote pili products and by-products.
3. *Human Resources*. Information on entities such as Pili growers, vendors, traders, processors alongside with their location and contact information and their products. Information on agencies is also being sought in this category.
4. *Research and Development*. This will be comprised of information on Pili related studies and researches. This includes the authors, publications, patents, and research products along with other related data.

2.5 Pili KM Ontology Design

For the Pili KMS to be semantic-ready, a Resource Description Framework (RDF) Model was prepared. According to W3C (World Wide Web Consortium), "RDF is a standard model for data interchange on the Web. It facilitates data merging even if underlying schemas differ, and it specifically supports the evolution of schemas over time without requiring all the consumers to be changed. [4]"

The RDF model is expressed using triples (Subject, Predicate, and Object) which composes the basic unit of information.

The Pili KMS' RDF, as shown in Fig. 3 has been modeled using sample facts to represent the correlation between Subjects and Objects. The Subjects were color-coded to represent the classes each of the facts belong.

Standards in creating ontologies were used in the semantic data. The following are consulted:

1. FOAF – acronym for "Friend of a Friend". This is used in creating a machine-readable ontology that describes persons, their activities, and their relation to other people and objects [5]. This ontology is used to pattern the Pili KMS's ontologies for the human resources (farmers, vendors, producers, and the like).
2. Dublin Core Schema [6] – a set of vocabulary terms used to describe digital resources (videos, images, web pages, etc.) as well as physical resources such as books and objects like artworks. The Dublin Core is used in this study to describe the subjects and objects that would represent materials like publications and research outputs.
3. AgroRDF [7] – is a schema that could represent and describe data on work farms including operating supplies like fertilizers, crops, and the lie. It covers topics to farm activities including "crop", "cropSpecies", etc. This can be used as a means of exchange data in a structured, standardized manner within the Agricultural Sector.
4. Plant Ontology [8] – A general reference ontology for Plant Sciences developed by Plant Ontology Consortium. This describes the anatomical structures and growth development of plant species.

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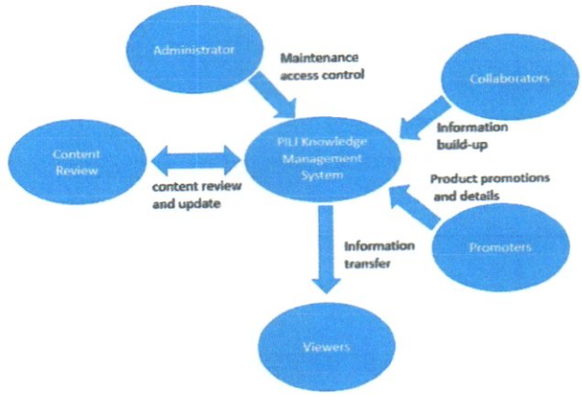


Figure 2: Pili KMS entities and their respective roles

The users of the system will be divided into different roles shown in Fig.2, namely:

1. *Administrator*. The administrator is responsible for the maintenance of the system. This entity is responsible for configuring users' access as well as the credentials of the users.
2. *Content Review*. This entity moderates and checks information for content validity before it will be published in the portal. They ensure that the information being posted is reliable and correct.
3. *Contributors (Collaborators and Promoters)*. These entities are responsible for the buildup of the knowledge related to Pili. Government Agencies, Agriculture Technicians, the Academes and Agriculture researchers are considered in this role. These entities are also responsible for the information update on Pili Human Resources such as on Growers, Vendors, Processors, and Traders.
4. *Viewers*. These entities are commonly the end-users. They can view information stored in the database and seek Pili related information. Their inputs are limited to only in participation in forum discussion.

2.3 System Development

The Pili Knowledge-base is developed using an Agile method called Scrum to be able to address the system's features (backlogs) and organize them according to priority. Time in scrum is partitioned into brief cadences called sprints. At each sprint, the team meets to see demonstrated shippable product increment and plan for the next step. This opens opportunities to expand the current scope of the Knowledge-base and address other potential data that could be related to access/retrieval of Pili-related data. This involves monitoring if the front-end (user interface) and the back-end (database) design are in harmony with each other. And, to monitor if the project runs according to defined schedules.

2.4 Database Design

The database of the Pili KMS is designed to cover, if not entirely, all possible aspects it needs to cater to. It is also modeled to handle massive access and retrieval of data. The database is normalized to handle complex queries needed to generate requested information. The tables in the database are categorized into four (4) main classes, namely:

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are written in a special language, the most commonly used is the Web Ontology Language (OWL). OWL documents can be published on the web and may refer to or be referred from other ontologies.

2 METHODOLOGY

2.1 Requirement Analyses and Scoping

The initial phase in the development of the Pili Knowledge Management System (KMS) and is considered as the most important phase, are the requirement analyses and scope definition. It involves identifying the data, functions, and other requirements that the system solution must provide. In the context of the Pili KMS, this includes characterizing what information the system must be able to handle, what functionalities the web interface must provide, the security mechanism, and others. Structured interviews, secondary data gathering, sampling of forms, and records are the techniques that are used in this phase.

2.2 Design Concept

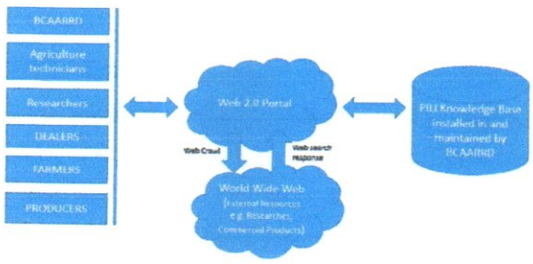


Figure 1: Architectural design of the information query and access of the Pili KMS

The system is designed to be a framework of knowledge sharing and retrieval among various collaborating entities responsible for promoting, cultivating, and managing Pili information. As shown in Fig. 1, entities BCAARRD (Bicol Consortium for Agriculture, Aquatic and Natural Resources Research and Development), Agriculture Technician, Researches, Dealers, Farmers, and Producers are part of the knowledge build-up and access through a collaborative interface (Web 2.0). Data and information gathered are stored in a structured centralized repository. Information not available in the repository is crawled/queried using an integrated Web 3.0 structure such as information like soil properties and climate conditions. Integrated Web 3.0 technology can also query independent research articles and contents related to Pili which are not captured by the collaborative knowledge buildup.

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reason calls the attention of creating and implementing methods and ways of pooling information on Pili into one readily accessible source.

Commercialization of Pili products, though to some extent has been a success, is yet to be backed by readily accessible data. Though Pili products are everywhere these days, there is no comprehensive and easily retrievable knowledge in this particular industry. It is believed that making information and data on Pili publicly available will enliven and invigorate the stakeholders in the industry and help boost the yield/ production, encourage more research and development efforts, promote innovations, step up marketability of various Pili-related products and consequently improve the living conditions of marginalized farmers and vendors.

Awareness on the part of farmers and vendors, on the marketability and viability of investing in the pili industry and rebound to a vibrant area of this agricultural sector. Moreover, scientists and researchers will be given access to a comprehensive source of information about Pili ranging from its classification, morphological characteristics, stress and diseases, production sites, uses/applications, products, production data, plantations, technologies, the profile of the farmers, vendors, and growers, prevailing market condition and other information.

1.3 Collaborative Web

The integration of Web 2.0 technologies into the design of the Pili Knowledge Management proposed will open opportunities for collaborative access to knowledge base similar to social media interaction and virtual communities. Farmers, Vendors, Researchers, and other stakeholders and the community as a whole will enable them to contribute content into the knowledge base through their accounts, creating a participatory environment. This collaborative environment is known to provide a culture of motivation and sparks interests among key players in the cultivation, production, and commercialization to adapt and move to the pace of what the world market needs.

1.4 Semantic Web

Though Pili Knowledge Base is built from its collaborative nature, information not available in the pool needs to be accessed somewhere else on the web. Agricultural processes such as Pili are dependent upon interlinked bodies of knowledge. [2]. The yield of the harvest does not just rely on the crop species but also the soil composition, local climate, and so forth. Also stated in [2] is the improbability that a solitary knowledge base will encompass all information that would be useful for the farmer. Isolated details stored on different domains would likely to happen. These isolated bits of knowledge can be aligned using semantic web technology which allows things to be queried as a bigger interwoven resource of knowledge. Semantic web technology has the potential to provide a shared structured representation of information gathered from non-real-time sources such as producer and retail systems and a real-time (immediate) data source such as sensor data readings.

To represent the knowledge within the semantic web application, ontologies are used by describing the types of classes of entities and by defining the relationships among them. Ontologies make implicit information about the content explicit allowing them to discover common meanings in diverse datasets. Thus, ontologies are an important component and form the foundation of the semantic web [3]. Ontologies

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Knowledge Base. Applications such as GraphsDB, Protégé, and SPARQL were used to model the semantic-side of the Pili Knowledge Management System.

CCS CONCEPTS

• **Web Technologies** → Web 2.0, Web 3.0 • **Database Management System** → Structured Relational Database Model • **Information System** → Knowledge Management Systems

KEYWORDS

Semantic Web, Collaborative Web, RDF, Ontology

1 INTRODUCTION

1.1 Pili (Canarium ovatum Engl.) Industry

Agricultural Researches have traditionally focused on staples and gives little attention to minor or underutilized crops [1]. Food supplying agencies and entities have been focused on a few staple crops being produced like rice and corn. Underutilized crops are sometimes being underestimated for their contribution. Most of the time crops considered to be neglected at the global level are staples at the regional level or even national level.

Pili (Canarium Ovatum Engl.) is one of the crops considered to be a minority. Studies related to Pili faced some downfalls due to limited information available. Another reason is that knowledge on the utilization of the crop is scattered and information about the germplasm is not readily accessible and is commonly found in grey literature or written in a little known language. Pili, on its current state, established limited existing knowledge on its genetic potential. These resulted in an inefficient approach to its genetic conservation and utilization.

Pili is one of the important fruit trees that bear edible nut found in the center of the diversity among species in the Philippines. Pili is considered as "tree of hope" and is highly appreciated because of its economic uses from sap to roots. It is hoped that soon, Pili will be ranked equal with cashew and macadamia in the world market [1].

The Philippine government had established initiatives in promoting and cultivating the survival of the Pili gene pool. Regular seminars are held on Pili production to interest farmers; the availability of seedlings to farmers of reasonable price and high-quality propagation materials; agricultural agencies also conducted regular visits on farmer's orchard by expert personnel and provide technical assistance to them and; a promise of Government assistance to Pili farmers in marketing their produce.

1.2 Pili Knowledge Management

Since information dissemination and its availability on Pili is lacking, limited, and scattered, the development and advancement to the knowledge and its utilization have been greatly hampered. This

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A Framework for Applying Semantic Web in the Design and Development of Web Portals for Agricultural Crops: The Pili (Canarium Ovatum Engl.) Case

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ABSTRACT- Discussed in this paper is the design and development of Pili (Canarium ovatum Engl.) Knowledge Base which was set up to be readily available for Semantic Web access. Though Pili is considered a minority crop, it is one of the important fruit trees found in the Philippines and is considered a "tree of hope" because of its usefulness from sap to roots. Information regarding Pili as of today are scattered and its availability is lacking affecting the Pili industry in general. To provide an intervention regarding the matter, it was deemed a necessity to create a central repository of Pili knowledge. This repository will establish a collaborative environment for the knowledge build-up by utilizing the Social Web (Web 2.0) technologies. This also builds a foundation for access to the Semantic Web (Web 3.0) whenever there is a need to access information not available in the repository. Ontology Standards were applied to ensure interoperability with other semantic websites. The development of the Pili Knowledge Base used Structured Relational Database Model that produced 152 normalized tables. The Semantic Web applied the Resource Description Framework to model the triples that reflects the current setup of the

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